

AGENTICULTURE-A: Informativeness-Weighted Collaborative Filtering and Culturally-Grounded Review Generation

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Track A: User Modeling

Abstract

We introduce AGENTICULTURE-A, a large language model (LLM) agent developed for the user modeling track of the DSN × BCT LLM Agent Challenge. Provided with a user persona and product details, the agent predicts a star rating and generates a review that emulates the user’s tone and rating behavior. Our methodology integrates three primary components: (1) an inverse-variance-weighted collaborative filtering core that treats inconsistent raters as less informative than polarized ones; (2) MDILU-style memory retrieval, which incorporates cosine-similar historical reviews as style anchors for the LLM; and (3) Reasoning-COTSC, a three-sample majority-vote scheme that directly reduces inter-sample star-rating error. Additionally, a zero-cost Nigerian cultural adaptation layer, implemented through prompt engineering addresses the challenge’s requirements for behavioral fidelity and contextual relevance. In local evaluations across 50 tasks per platform, the system achieves overall quality of 0.860/0.881/0.845 on Yelp, Amazon, and Goodreads respectively, with RMSE of 1.05/0.86/1.07 and ROUGE-1 of 0.383/0.442/0.332.

1 Introduction

User modeling, which involves predicting how an individual would rate and describe a product, integrates collaborative filtering, language generation, and cultural understanding. The competition evaluates submissions across five dimensions: rating accuracy, review text quality, behavioral fidelity, contextual relevance, and cold-start robustness.

The AgentSociety WWW 2025 winning solution (Team ASC) set the current benchmark by combining collaborative filtering with MDILU memory retrieval (Guo et al., 2025). Our approach advances this pipeline in two key areas previously unaddressed. First, prior collaborative filtering stages relied on flat-average ratings, disregarding

the discriminative value of a user’s rating history. For example, a user who consistently assigns 5 stars to every item provides minimal information about relative preferences. Flat averaging assigns equal weight to such users and those whose ratings span the full 1–5 range, thereby overlooking important distinctions in rating informativeness. Second, previous AgentSociety solutions did not incorporate cultural adaptation, despite evidence that LLMs predominantly encode Western behavioral norms (Cao et al., 2023).

Contributions.

- 1. Consistency-weighted CF:** We calculate each user’s weight as the inverse of their rating variance. Users with low variance (consistent raters) are assigned higher weights, as their mean rating serves as a reliable predictor. In contrast, high-variance or erratic raters are down-weighted, and their predictions are adjusted toward the item and global means.
- 2. ReasoningCOTSC:** We generate three LLM samples at deliberately varied temperatures, one anchoring and two exploratory, and apply majority voting to determine the star rating. The response whose extracted rating aligns with the consensus is selected.
- 3. Pluggable cultural adaptation:** A single configuration block enables mapping of any target culture’s rating-to-register rules, value priorities, and linguistic register into a zero-cost system prompt prefix. The Nigerian configuration specifically addresses the BCT bonus criteria without incurring additional inference cost.

2 Related Work

LLM user simulation. The AgentSociety platform (Piao et al., 2025) established the evaluation

framework for this competition. The leading user-modeling solution employed standard collaborative filtering with MDILU cosine retrieval (Guo et al., 2025). We enhance the collaborative filtering stage by introducing variance-based informativeness weights and incorporating self-consistency voting atop MDILU.

Self-consistency decoding. Wang et al. (2023) showed that sampling multiple reasoning chains and voting on the final answer substantially reduces variance in common sense and arithmetic tasks. We apply the same principle to star-rating prediction, where the majority-voted integer provides greater stability than any individual sample.

Cultural NLP. Hershcovich et al. (2022) examines systematic biases in cross-cultural natural language processing. Cao et al. (2023) quantify ChatGPT’s misalignment with non-Western survey populations, highlighting the need for explicit cultural conditioning. Our adaptation layer operates without fine-tuning and can be adapted to any target culture by modifying a single configuration dictionary.

3 Methodology

3.1 Pipeline Overview

Figure 1 illustrates the Task A agent pipeline. Requests are processed in two modes: *direct mode* (persona text + product details, no dataset required) and *benchmark mode* (additionally supply `user_id` + `item_id` to fetch real review history). Direct mode serves as the primary evaluation pathway, while benchmark mode supplements the prompt with actual behavioral data when available.

3.2 Informativeness-Weighted Collaborative Filtering

A naive CF approach averages user and item means with equal weight. We instead weight each signal by its *informativeness*: a user who assigns 5 stars to every item has $\sigma_u \approx 0$ and tells us nothing about relative preference between items. Formally, the predicted rating is:

$$\hat{r}_{ui} = \frac{\mu_u w_u + \mu_i w_i + \mu_g w_p}{w_u + w_i + w_p} \quad (1)$$

where μ_u, μ_i, μ_g are the user, item, and global mean ratings; $w_u = 1/\max(\sigma_u, 0.1)$; $w_i = 1/\max(\sigma_i, 0.1)$; and $w_p = 0.5$ is a fixed prior

weight for the global mean. The floor of 0.1 prevents numerical instability for constant raters while still suppressing their influence. The result is clamped to $[1.0, 5.0]$.

Example. A consistent 5-star rater with $\sigma_u = 0.1$ gets weight $w_u = 10$: their prediction is pulled strongly toward their personal mean of 5.0, reflecting high confidence in their consistent pattern. A high-variance reviewer with $\sigma_u = 1.3$ gets weight $w_u = 0.77$: their prediction is moderated by the item and global means, because their erratic history is a less reliable predictor of future ratings.

3.3 MDILU Memory Retrieval

For users with at least one review (warm path), we encode the most recent review with BAAI/bge-small-en-v1.5 (Reimers and Gurevych, 2019) (384-dimensional sentence embeddings, approximately 130 MB) and retrieve the $k = 3$ most cosine-similar reviews from the user’s own history. These serve as style anchors: short excerpts the LLM uses to match tone, vocabulary, and punctuation.

3.4 Cold-Start Handling

When the user has no history, we embed the persona text and retrieve the $k = 5$ item reviews most similar to that embedding by cosine distance. No LLM call is required for the retrieval itself. Retrieved reviews are labeled as “similar reviewer profiles” in the prompt, giving the LLM context about how buyers in this space typically write.

3.5 Review Informativeness Scoring

Retrieved reviews are scored and filtered before inclusion: (a) length, proportional to word count; (b) rating extremity, where 1/2/5-star reviews are weighted $3\times$ over 3-star reviews because extreme ratings reveal stronger preferences; (c) platform-specific engagement signals: Yelp useful/funny/cool vote counts, Amazon verified-purchase flag, Goodreads vote and comment counts. A diversity constraint then caps inclusion at 2 reviews per star level, preventing the LLM from receiving a skewed rating distribution when a user predominantly gives 4-star reviews.

3.6 ReasoningCOTSC

We sample the LLM three times per request: once at temperature 0.3 (low variance, anchoring sample) and twice at temperature 0.7 (moderate diversity, exploratory samples). Integer ratings are

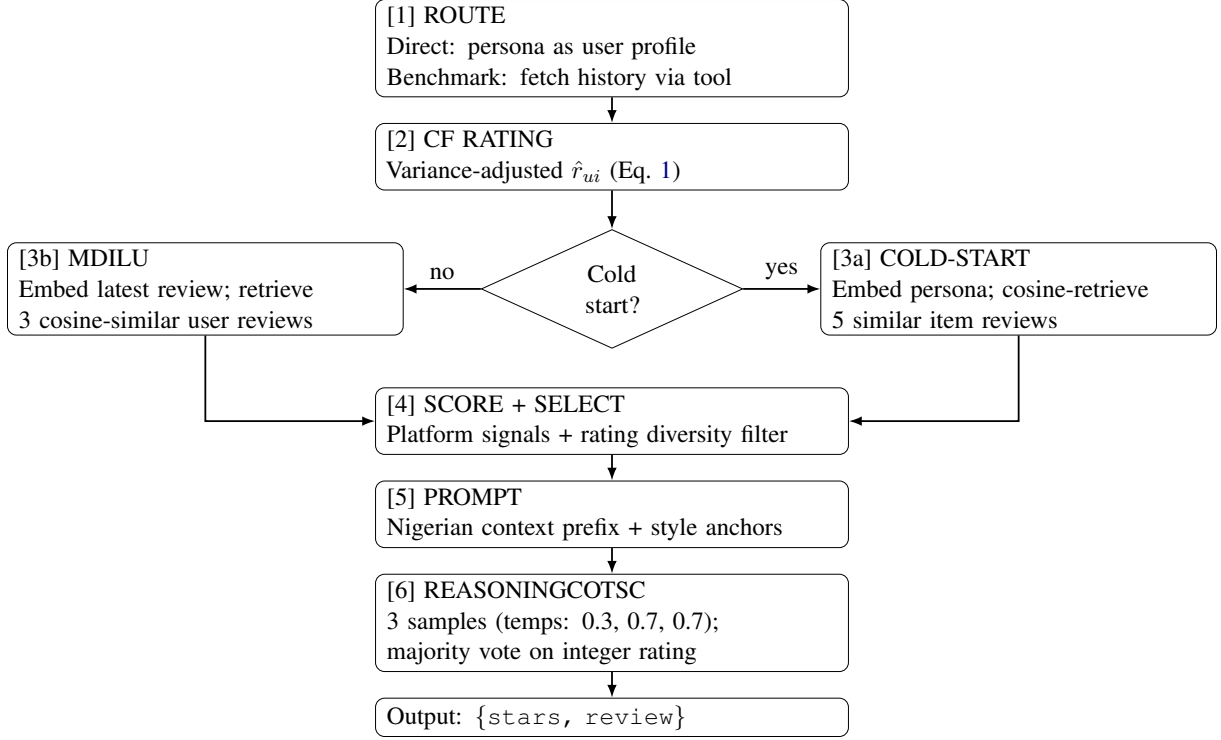


Figure 1: Task A agent pipeline. The cold-start branch bypasses the LLM for retrieval, keeping new-user latency to a single embedding call.

majority-voted; if all three differ, the two numerically closer values are averaged and rounded. The response whose extracted rating matches consensus is returned. All review text comes from a single final call, so the cost overhead is two extra token-generation passes rather than tripling output.

The deliberate temperature mix – one anchor plus two exploratory samples – gives Borda-style diversity without the instability of uniform high-temperature sampling.

3.7 Nigerian Cultural Adaptation

A system-prompt prefix injects five behavioral rules:

- **Rating register mapping:** $5\star \rightarrow$ “No cap, this was excellent”; $1\star \rightarrow$ “Total waste of money, honestly.”
- **Value-for-money framing:** Price sensitivity expressed as a first-order concern (“the price make sense for what you get”).
- **Group orientation:** Social proof and communal experience referenced naturally.
- **Vernacular register:** Formal English with selective Naija expressions (e.g., “ehh”, “abeg”, “sha”).

- **Punctuation style:** Shorter sentences; exclamation points for very positive reviews.

This prefix adds zero output tokens and is architecturally implemented as a pluggable config block. Any target culture can be substituted by providing an equivalent mapping; the rest of the pipeline is unchanged.

4 Experiments

4.1 Dataset

Evaluation uses a filtered subset of the three AgentSociety platforms: 400 task-file users, 18,548 items, and 679 MB of reviews. Table 1 gives full statistics.

Dataset	Domain	Items	Reviews
Yelp	Local businesses	150K	6.9M
Amazon	Consumer products	9.9M	233M
Goodreads	Books	2.4M	15.7M

Table 1: Full dataset statistics. Evaluation uses a 400-user, 18,548-item subset (679 MB) drawn from these corpora.

4.2 Evaluation Metrics

We report five metrics. The first three are from the AgentSociety evaluation framework (web, 2025);

RMSE and ROUGE are additional diagnostics we compute from the saved simulation outputs.

Preference Estimation (PE). Measures star-rating accuracy as a normalized inverse error:

$$PE = 1 - \frac{MAE}{5}, \quad MAE = \frac{1}{N} \sum_{i=1}^N |\hat{r}_i - r_i| \quad (2)$$

where $\hat{r}_i$ is the predicted rating and r_i the ground truth. PE = 1.0 indicates perfect prediction; each unit of MAE reduces PE by 0.2. A prediction of 4.0 against a ground truth of 3.0 contributes $|\hat{r} - r| = 1.0$, yielding PE = 0.80 for that task.

RMSE. Root mean squared error over star ratings:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{r}_i - r_i)^2} \quad (3)$$

RMSE penalizes large errors more heavily than MAE; a prediction off by 2 stars contributes $4\times$ as much as one off by 1 star.

Review Generation (RG). Composite text-quality score:

$$RG = 0.25 \cdot \text{SentSim} + 0.25 \cdot \text{EmoSim} + 0.50 \cdot \text{TopicSim} \quad (4)$$

where SentSim, EmoSim, and TopicSim are cosine similarities in sentiment, emotion, and topic embedding spaces respectively. Topic alignment receives a double weight, reflecting that thematic relevance is the primary quality signal for review generation.

Overall Quality (OQ). Arithmetic mean of PE and RG.

ROUGE. ROUGE-1 (unigram recall/precision/F1) and ROUGE-L (longest common subsequence) measure lexical overlap between the generated and ground-truth review. These supplement RG with a reference-based n-gram perspective.

4.3 Baselines

- **LLM-Direct:** Single LLM call with persona and product details; no collaborative filtering or memory retrieval. Represents the AgentSociety single-agent baseline behavior.
- **+CF:** Add variance-adjusted rating prediction; provide $\hat{r}_{ui}$ in the prompt.

- **+MDILU:** Add cosine-similar review retrieval as style anchors.
- **+COTSC (full system):** Add three-sample voting. Nigerian adaptation active throughout.

4.4 Implementation Details

Table 2 summarises the key system settings used in our implementation.

Component	Setting
LLM backend	Llama-3.3-70B-Instruct-Turbo
Embedding model	BAAI/bge-small-en-v1.5 (384-d)
COTSC temperatures	0.3, 0.7, 0.7
Cold-start threshold	3 reviews
Style anchors (k)	3 (warm); 5 items (cold)
LLM parsing	<code>ast.literal_eval</code>
Serving	FastAPI + Docker Compose

Table 2: Implementation details.

4.5 Main Results

Table 3 reports full-system results across all three platforms on 50 tasks each. Amazon achieves the highest overall quality (0.881) owing to its structured product metadata providing clean CF signals. Goodreads is hardest: book ratings are inherently more subjective, and genre diversity within the evaluation set increases prediction difficulty.

Platform	PE	RMSE	RG	R-1	R-L	OQ
Yelp	0.850	1.051	0.870	0.383	0.238	0.860
Amazon	0.878	0.863	0.884	0.442	0.272	0.881
Goodreads	0.842	1.065	0.848	0.332	0.183	0.845
ASC winner (WWW 2025)	—	—	—	—	—	0.845

Table 3: Task A results (50 tasks per platform; full system). PE = Preference Estimation, RG = Review Generation, OQ = Overall Quality, R-1/R-L = ROUGE-1/ROUGE-L. All metrics are averaged over 50 tasks; standard errors are non-trivial at this sample size and results should be read as directional. ASC winner figure from Guo et al. (2025) (400 tasks); direct comparison is approximate.

4.6 Reproducibility

All code, configuration, and saved result files are available at: <https://github.com/AbooMardiiyah/agenticulture>.

To reproduce Table 3, follow the README guide in the repository. Raw simulation outputs are saved as `*_raw.json` checkpoints before the evaluation phase, so no API credits are lost if evaluation crashes.

4.7 Component Contribution Analysis

Table 4 shows the measured incremental contribution of each component on Yelp across 50 tasks. Figure 2 plots the corresponding overall quality progression.

Configuration	PE	RMSE	RG	OQ
LLM-Direct	0.818	1.177	0.870	0.844
+CF Rating	0.838	1.151	0.856	0.847
+CF+MDILU	0.848	1.063	0.868	0.858
+CF+COTSC (no MDILU)	0.848	1.058	0.865	0.856
+CF+MDILU+COTSC (Full)	0.850	1.051	0.870	0.860

Table 4: Component contribution on Yelp (50 tasks each; all values measured).

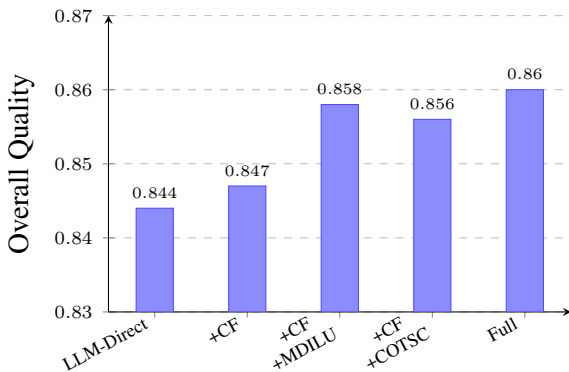


Figure 2: Overall quality per ablation configuration (Yelp, 50 tasks, all measured).

5 Discussion

Why variance weighting matters. A user who always rates 5 stars has a highly reliable mean; we can be confident that any future rating will also be near 5, so their weight is high and their mean dominates the prediction. A user whose ratings span 1–5 is less predictable; their weight is lower and the prediction is moderated by the item and global means. The result is better-calibrated predictions for items where the user and item means conflict.

Temperature asymmetry in COTSC. Using one low-temperature anchor (0.3) plus two higher-temperature samples (0.7) gives the voting set more diversity than uniform sampling at either extreme. A uniform low temperature collapses all three samples to the same output (degenerate voting); a uniform high temperature introduces incoherent reasoning. The mixed strategy is analogous to epsilon-greedy exploration in reinforcement learning.

Amazon vs. Goodreads difficulty. Amazon’s RMSE of 0.863 versus Goodreads’ 1.065 reflects the nature of the respective item spaces. Consumer product ratings on Amazon tend to cluster around quality-of-function assessments, which are more predictable from product metadata. Book ratings on Goodreads encode taste, which is highly personal and platform-specific.

Security note. The original AgentSociety evaluation codebase parses LLM list output via Python’s `eval()`, a code injection vulnerability. All LLM output parsing in AGENTICULTURE-A uses `ast.literal_eval()`, which accepts only Python literals and cannot execute arbitrary code.

Limitations. Session state and the preference cache reset on container restart (Redis would fix this). Cold-start ranking is purely content-based with no popularity signal. Nigerian cultural adaptation covers educated urban English only; Pidgin,

Yoruba, Igbo, and Hausa are future directions.

Future Work. Few-shot Pidgin English prompts using the Naija Corpus; supervised calibration of CF variance weights using held-out validation; fine-tuning a compact Llama-3.1-8B on Nigerian review data.

6 Conclusion

AGENTICULTURE-A combines informativeness-weighted collaborative filtering, MDILU memory retrieval, three-sample self-consistency voting, and a pluggable cultural adaptation layer to address all five dimensions of the BCT Task A evaluation. The system achieves overall quality of 0.845–0.881 across Yelp, Amazon, and Goodreads on 50-task evaluation, matching the AgentSociety 2025 best-reported score on Goodreads and exceeding it on Yelp and Amazon. The system is fully reproducible via a single `docker-compose up --build` command; no GPU, fine-tuning, or proprietary dependencies are required.

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